



Armada Kalamunda Group

Neurovascular Assessment and Observation Guideline

1. Purpose

This document guides clinical staff in neurovascular assessment and observation and will be undertaken when there is any concern of acute neurovascular compromise to a limb or digit.

An initial set of observations will be documented and routinely repeated at four hourly intervals as a minimum. The frequency of observations will be increased where the risk to neurovascular compromise is increased, or on the orders of a medical officer.

Inform the Shift Coordinator and Medical Officer if any results deviate from baseline observations

2. Guidelines

2.1 Peripheral vascular assessment

Always consider the risk of compartment syndrome using the 7 P's – see Appendix 2.

2.1.1 Pain

- Assess the patients' pain score at rest, during active/passive movement, pre and post analgesic medication administration.
- Ask the patient to rate their pain using the numerical pain scale 0-10. If the patient is unable to self-report pain scores it is important that they are assessed for non-verbal pain behaviours.
- If pain is un-retractable and inconsistent with the injury or mechanism of injury, then urgent assessment for compartment syndrome should be initiated.

2.1.2 Colour

- Visually inspect the patient's limb/digit and document any changes in colour.

2.1.3 Temperature

- Assess the temperature of the whole limb/digit using the back of the hand. If an injury is present, assess the temperature proximal and distal to the site of the injury.

2.1.4 Capillary refill

- Capillary refill should be used to measure arterial perfusion in patients who present with an inaccessible pulse due to plaster casts/bandages etc.

 Pressure is applied to nail bed until it turns white	1. Remove any nail polish and ensure the nail is clean 2. Press the nail bed downward with light to moderate pressure for 5 seconds whilst the digit is at heart level until blanching occurs, release the pressure and observe how long it takes for normal colour to return 3. If the nail is damaged or unsuitable for assessment use the skin area below the nail <i>Note: Pressure should not need to be applied for any longer than five seconds. Consideration should be given to environmental temperature when assessing capillary refill performance</i>
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2.1.5 Distal pulses

- Distal pulses are assessed to determine the presence of arterial blood flow so as to ensure there is adequate perfusion reaching past the level of injury/surgery and the remainder of the limb effectively.
- Document pulses as - Absent, Weak, Strong or Doppler.

Pulses that may need to be assessed:

Upper Limb

- Brachial
- Radial

Lower Limb

- Femoral
- Popliteal
- Posterior Tibialis
- Dorsalis Pedis

Note. The dorsalis pedis can be absent in 12% of the population. Use posterior tibialis pulse to establish arterial perfusion if absent. Both pulses should be assessed if possible.

If the pulses are deemed weak or absent, then the pulse should be assessed using a Doppler.

3. Doppler auscultation

A Doppler machine can be used to detect and hear pulses that are otherwise difficult to find.

3.1 Procedure

- Clean the end of probe before and after use. Alcohol swabs can be used on the probe tip and for general cleaning purposes.
- Use a liberal amount of contact gel between skin and probe (use water-based ultrasound gel only).
- Gently place the probe at a 45 degree position to the skin surface and adjust position until the loudest audio signal is obtained. The audible volume can be adjusted. Applying too much pressure to the probe can occlude the artery and make it difficult to auscultate the pulse. Arteries emit a high pitched pulsatile sound. Veins emit a non-pulsatile sound similar to rushing wind.
- Once the optimum position is found keep the probe as still as possible.

- Do not use in the presence of flammable gases such as anaesthetic agents, oxygen or Entonox.
- Mark the pulse area once it is found on the patient's limb.

3.2 Peripheral Neurological Integrity

- Peripheral neurologic integrity is determined by sensation and motor function of the upper and lower extremities.

3.3 Sensation

- An assessment of sensation should be made by first asking the patient if he or she feels any altered sensation of the affected limb – consider any nerve blocks or epidurals. The patient should be asked to close their eyes while the sensation is being evaluated. A patient should be able to detect sensation if the affected digits are touched lightly. If the patient complains of pins and needles or paraesthesia further investigation is warranted.

3.4 Motion

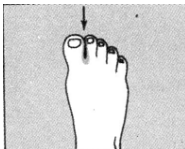
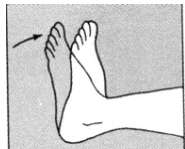
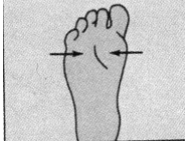
Assess active or passive range of movement in both limbs:

- First the unaffected
- Then the affected side

Paralysis of a limb is to be described as:

- Present
- Decreased
- Absent

Taking into consideration the restrictions of dressing and or plaster and their injury, the patient should be able to demonstrate active movement of the joint. Severe pain on passive movement of digits or toes requires immediate attention, as it may be a sign of compartment syndrome

Peroneal Nerve	 Sensation Feeling pressure at the web space between the great toe and second toe	 Motion The patient is able to dorsiflex the ankle and extend toes at the metatarsal phalangeal joint
Tibial Nerve	 Sensation Feeling present on the medial and lateral surfaces of the foot	 Motion Have the patient plantar flex ankle and toes

Radial Nerve	 Sensation Feeling at the web space between the thumb and index	 Motion Have patient hyperextend thumb then wrist and hyperextend the four fingers at the metacarpal phalangeal joints.
Ulnar Nerve	 Sensation Touch the distal fat pad of the small finger	 Motion Have patient abduct all fingers
Median Nerve	 Sensation Feeling at the distal surface of the index finger	 Motion Have patient oppose thumb

Findings	Normal Range	Inform MO
Pain	Adequately Relieved	Increasing/Unrelieved
Skin Colour	Pink	Pale, Mottled or Cyanotic
Temperature	Warm/Cool	Hot/Cold
Capillary Refill	<2 Seconds	>2 seconds
Distal pulses	Strong/Weak	Absent/Doppler
Sensation	Normal	Absent/Impaired
Motion	Normal	Absent/Impaired

4. Equipment required

AKMR 146.1 Neurovascular Observation Chart.

5. Documentation

- Document all findings on the AKMR146.1 or department specific forms and patient notes.
- If there are any barriers to completing the Neurovascular Assessment Observation fully e.g. plasters or back slabs insitu, ensure this is clearly documented.

6. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

7. Supporting information

The following documents support the implementation of this policy:

- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)

8. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 6 – Action 6.6 Correct identification and procedure matching
- The health service organisation specifies the:
 - a. Processes to correctly match patients to their care
 - b. Information that should be documented about the process of correctly matching patients to their intended care
- Standard 6 – Action 6.7 Clinical handover
- The health service organisation, in collaboration with clinicians, defines the:
 - a. Minimum information content to be communicated at clinical handover, based on best-practice guidelines
 - b. Risks relevant to the service context and the particular needs of patients, carers and families
 - c. Clinicians who are involved in the clinical handover
- Standard 8 – Action 8.3 Partnering with consumers
- Clinicians use organisational processes from the Partnering with Consumers Standard when recognising and responding to acute deterioration to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 8 – Action 8.4 Recognising acute deterioration
- The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:
 - a. Document individualised vital sign monitoring plans
 - b. Monitor patients as required by their individualised monitoring plan

- c. Graphically document and track changes in agreed observations to detect acute deterioration over time, as appropriate for the patient
- Standard 8 – Action 8.6 Escalating care
- The health service organisation has protocols that specify criteria for escalating care, including:
 - a. Agreed vital sign parameters and other indicators of physiological deterioration
 - b. Agreed indicators of deterioration in mental state
 - c. Agreed parameters and other indicators for calling emergency assistance
 - d. Patient pain or distress that is not able to be managed using available treatment
 - e. Worry or concern in members of the workforce, patients, carers and families about acute deterioration

10. Document owner

Director Nursing and Midwifery

Enquiries relating to this guideline may be directed to:

Title: Coordinator Nursing
 Department: Hospital wide
 Email: shane.hastings@health.wa.gov.au

11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V5	07/08/2023	07/08/2026	Scheduled review

12. Authorisation

This guideline has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Helen Fullarton – Director Nursing and Midwifery
Authorisation date	04/08/2023
Published date	07/08/2023

13. Appendix 1: Procedural guideline

- Key points prior to commencing any procedure
- Consider cultural, ethical and communication requirements.

Review patient history and diagnosis for clinical conditions, medications and psychosocial factors that could influence observations.

- Refer to previous observation parameters if available for comparison.
- Explain the procedure/s to the patient, family and/or carer and gain appropriate consent.
- Maintain standard safety precautions for infection control.
- Ensure that the unaffected limb is assessed concurrently with the affected limb during the neurovascular assessment.
- Ensure that there is a full assessment made of the patients' pre morbid conditions and any local factors that may affect the patients' neurovascular assessment.

General Information

Neurovascular assessment involves the evaluation of the peripheral circulation, vascular integrity of a limb and neurological status of the patient. Neurovascular impairment is frequently caused by pressure on the nerve or on the vascular supply to an extremity where prompt recognition can minimise the risk of complications to the patient.

Some conditions and factors can affect the reliability of the assessment e.g. paralysis, diabetes, peripheral vascular disease, epidural infusion, dressing materials and patient limb positioning.

If the clinicians' ability to palpate peripheral pulses or to observe for oedema or movement is hindered, it is sufficient to rely on the remaining parameters for assessment.

Indications for Procedure:

- Trauma to an extremity e.g Fractures/Dislocations/Crush Injury/Deep Lacerations
- Application of Traction, Plaster, Back Slab, restrictive dressings, splints or casts
- Post Operatively
- Spinal injuries
- Vascular injuries
- Burns Patients
- Any report of altered sensation
- Underlying co morbidities i.e. tumours
- Tendon, artery, nerve repairs or split skin grafts
- Some cognitively impaired and/or uncooperative patients

14. Appendix 2: Compartment syndrome

Compartment syndrome is a potential complication of musculoskeletal trauma and surgery resulting from increased pressure in the muscle compartment in the closed fascial space, which results in cellular anoxia, muscle ischaemia and death. Regular Neurovascular Observation Assessment must occur to allow for early recognition of abnormal findings. Early recognition is paramount to facilitate treatment to reduce severity of the compartment syndrome and the possibility of limb loss or death if left untreated.

Signs of Compartment Syndrome

Assessment for the signs and symptoms of Neurovascular deficit should take into account the following '7Ps':

Pain	Excessive pain on passive extension and flexion of the extremity. Excessive pain to what is expected that is not relieved by increasing amounts of analgesia
Pallor (colour)	May indicate potential late sign of arterial injury
Polikilothermia (temperature)	Coldness can indicate inadequate arterial blood flow or hot as inadequate venous return
Pulselessness	No pulse present can indicate death of a tissue
Paraesthesia (sensation)	Abnormal sensations e.g. numbness, tingling of extremity. Lack of ability to complete active movement, caused by prolonged nerve compression or nerve damage
Paralysis	Deficit may cause muscles in the affected compartment to become paralysed as a consequence to nerve damage or necrosis
Pressure	Skin appears tight and shiny